Local goodness-of-fit measures for neural decoding

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Abstract

State-space models are widely used in neuroscience to infer latent states from observations, but validating them remains difficult. Interpretable models are necessarily simplified relative to the data, and determining the level of complexity needed to describe latent variables accurately is challenging. This challenge results from two main factors. First, because latent variables are inherently unobservable, decoding quality cannot be validated directly against observed data. Second, prediction errors caused by model misspecification can be difficult to interpret. We therefore developed metrics designed to identify localized periods when individual observations conflict with the model's accumulated estimate. Each metric assesses the consistency between the information contained in an individual observation (here, a single spike) and the information summarized by a state distribution over a chosen window of data. Here, we use the one-step predictive distribution,

which carries forward information from all observations through the preceding time bin; more generally, the comparison could use filtering or smoothing distributions defined over other windows. The metrics fall into three categories: 1) measures of overlap between highest probability-density regions of the state distributions; 2) predictive checks based on the model-implied predictive probabilities of individual observations; and 3) the Kullback–Leibler (KL) divergence, included as a familiar baseline that quantifies difference rather than consistency. We applied these metrics to simulated data and to real rat hippocampal neural data, demonstrating their ability to identify periods of model misspecification and to provide interpretable feedback for model improvement. Together, the metrics provide complementary tools for neuroscientists to assess the local goodness-of-fit of state-space models, enabling more informed decisions about model refinement and interpretation of neural decoding results.

Keywords: Neural decoding, State-space models, Goodness-of-fit

Introduction

Internal brain states related to memory, emotion, and attention play central roles in behavior. They arise from a time-varying interplay between the external world and internal processes and are not directly observable, making them difficult to characterize.

Recordings of neural activity, including spike trains, calcium signals, and voltage traces, provide experimental measurements influenced by these latent states. State-space models make it possible to relate these measurements to estimates of the latent states themselves. Specifically, these models provide a method for relating complex, high-dimensional neural activity to estimates of low-dimensional latent states and have provided insights into neural representations and dynamics associated with taste [1], spatial location [2], and aggression [3].

State-space models aggregate information about the latent states over time. They combine the information from the data at the current time (the normalized likelihood distribution) with the accumulated information used to predict the latent state (the predictive distribution). This powerful

combination enables robust estimation of the latent state, effectively balancing current information with the accumulated history of the system and prior knowledge.

Despite the wide use of state-space models in neuroscience, methods to validate them remain underdeveloped and underused. When validation is skipped or reduced to a single pass/fail verdict, a model that fails in one behavioral period but succeeds elsewhere is accepted or rejected wholesale, and the specific assumption responsible for a misfit is never identified—so the model is never refined in a targeted way. One reason models are not refined this way is that latent states are inherently difficult to evaluate [4]: in the absence of ground truth, model validity can only be inferred indirectly, for example by comparing relative likelihoods across models, the quality of reconstructed signals, or performance on downstream tasks. Neural data are also dynamic, so model fit can vary substantially over time. Compounding this, the most common validation tools—cross-validation and "shuffle" permutation tests—summarize the entire recording in a single number: they report whether a model generalizes better than chance or better than a simpler alternative, but not *when* during a recording the model fits poorly or *which* of its components is responsible. Consequently, these methods can reject intentionally simplified yet scientifically meaningful models while offering little guidance for improving them. For example, analyses that decode rat movements from hippocampal place cells typically ignore known features of the rat's movement (such as momentum) and known features of neurons (such as refractoriness and bursting). These omissions can produce detectable model mismatch even when the resulting decoded trajectories remain scientifically informative [5–7].

State-space models already contain rich diagnostic information in the form of distributions used to estimate the state process at each time: the normalized likelihood, which reflects information from only the current neural observation, and the predictive distribution, which aggregates information from all past observations through the state-space model structure. One potential source of targeted, temporally local diagnostic information is a direct comparison of these two distributions. We do not expect them to be identical. The predictive distribution aggregates information from many past spikes, whereas the likelihood reflects only the current observation, so the two carry different amounts of information about the state and will typically differ in shape and spread.

What matters for model assessment is therefore not whether the two distributions differ, but whether they are *consistent*. We call them consistent when they concentrate their probability mass over overlapping regions of the state space, and inconsistent when those regions are largely disjoint. Consistency is distinct from accuracy: accuracy asks whether the decoded state matches an external ground truth, which is unavailable for a latent process, whereas consistency asks only whether the two internal sources of information about that state—the accumulated past and the current observation—point to the same region. Two distributions may be broad or narrow, and the decoded state may sit far from the animal's physical location, yet still be consistent as long as their high-probability regions overlap. When instead they provide inconsistent information, the mismatch can reveal where, when, and potentially how the model fails to capture the structure of the data. Examining these discrepancies therefore provides a direct, interpretable measure of model–data agreement—analogous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of local goodness-of-fit methods, each built on this comparison between the predictive distribution and the likelihood. The most familiar way to compare two distributions is to measure their divergence, and therefore we include Kullback–Leibler (KL) divergence as a natural point of reference. However, divergence measures quantify how much two distributions differ, not whether they are consistent, so they flag any mismatch, including one that arises from a broad predictive distribution or a narrow likelihood, even when the model captures the data structure well. We therefore develop two consistency-focused measures, highest probability-density (HPD) overlap and a rank-based predictive p-value derived from a predictive check, and show that they flag genuine model–data inconsistency and provide actionable feedback to neuroscientists. Using simulated and real hippocampal data, we demonstrate how each metric responds to different types of misspecification and argue that consistency is the right target for local goodness-of-fit assessment. This approach offers a path for neuroscientists to move from binary model rejection (is this model good or bad?) toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

Methods

State-space models

The state-space paradigm is particularly well suited for neural data analysis because many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the trajectories being replayed are not directly observable by experimentalists [8]. In such cases, state-space models are used to simultaneously estimate the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \ldots, t_K$, we define an unobserved state process x_k with initial state x_0, which influences a neural observation process y_k at each subsequent time t_k, $k = 1, \ldots, K$. In the models analyzed here, y_k comprises the spikes of each sorted unit in the kth time bin; in a clusterless model, each spike would instead be identified by its waveform features. We define a state-space model by combining a state-transition distribution $p(x_k \mid x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k \mid x_k)$, which defines the relationship between the neural activity and the state. To simplify the problem, we assume that the state-transition distribution depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Fig. 1a shows a schematic of this state-space model structure.

This state-space model structure allows for iterative estimation of the state process at each time step based on all the neural data up to that time (Fig. 1b). If the distribution of the state at time t_{k-1} has been computed, we can integrate it against the state-transition distribution to obtain a one-step predictive distribution, $p(x_k \mid y_{1:k-1}) = \int_{x_{k-1}} p(x_k \mid x_{k-1}) \cdot p(x_{k-1} \mid y_{1:k-1})\, dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current time step: $p(x_k \mid y_{1:k}) \propto p(y_k \mid x_k) \cdot p(x_k \mid y_{1:k-1})$.

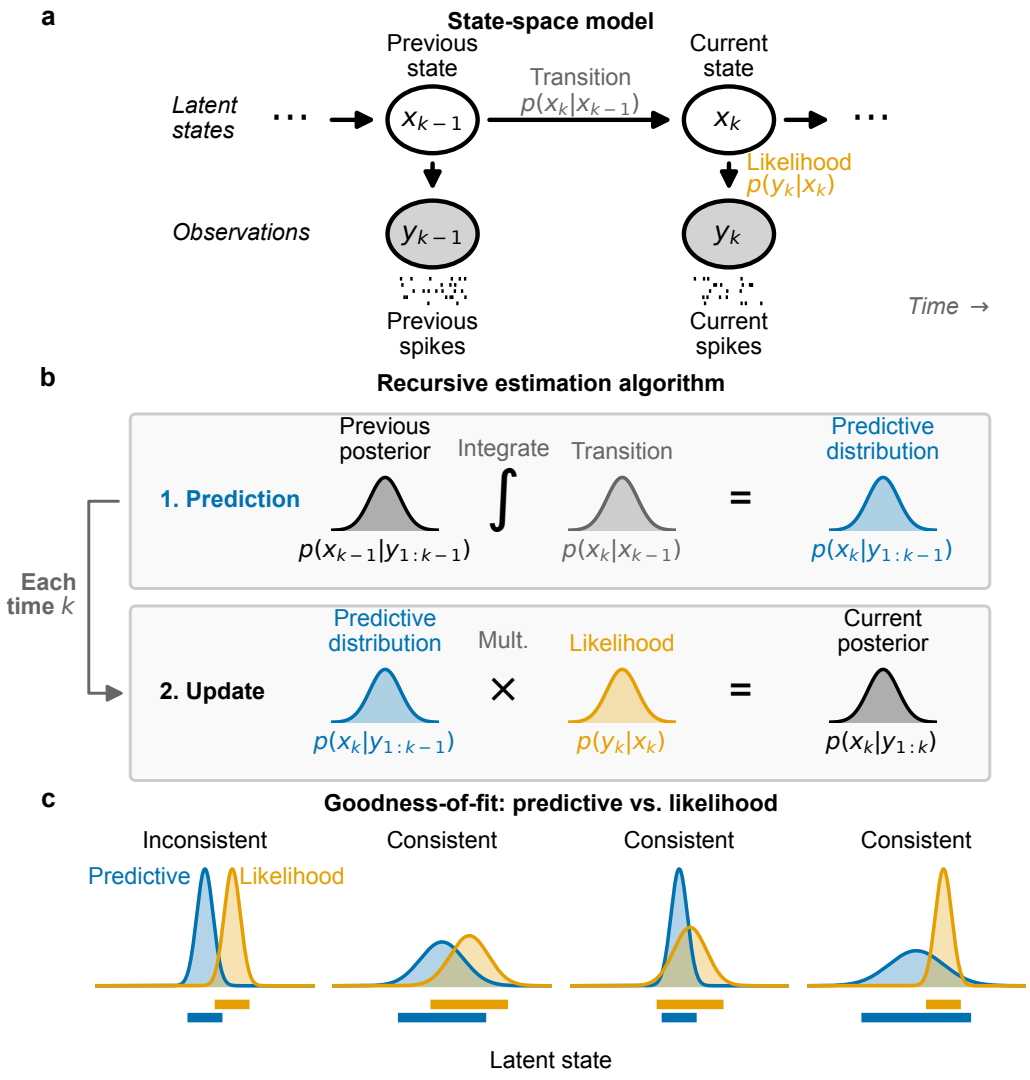


Fig 1. State-space model and local goodness-of-fit measures. **(a)** The graphical representation of the state-space model includes the state-transition model, $p(x_k \mid x_{k-1})$, and the observation likelihood, $p(y_k \mid x_k)$. **(b)** These model components are sufficient to compute an iterative expression for the filtering distribution at time t_k, $p(x_k \mid y_{1:k})$, from the distribution at the previous time step. This involves first integrating the previous filtering distribution against the state-transition distribution to compute the one-step predictive distribution, $p(x_k \mid y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. **(c)** For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and the normalized likelihood of that spike. Bars beneath each pair mark the high-probability region of each distribution. We consider the distributions inconsistent when they assign high probability to mostly disjoint subsets of the state space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

Common approaches for assessing the goodness-of-fit of state-space models include examining cross-validated error measures in the state process when the state is observable [9], cross-validated error measures in the observation process such as the mean-squared prediction error [10], and the distribution of prediction residuals [11]. Although these approaches can be useful, they may be difficult to interpret for neural data because intentionally simplified state-space models often exhibit some detectable lack of fit even when they remain scientifically useful.

Here, we take a different approach that relies on a set of local goodness-of-fit measures. Each measure compares the one-step predictive distribution with either the normalized likelihood of a single spike as a function of state or the spike itself. These measures therefore consider the consistency between information accumulated through the state-space model over past spiking and the information from the most recent spiking activity. Critically, we do not expect these distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we use the operational notion of consistency illustrated in Fig. 1c. We consider the distributions inconsistent when their probability masses are concentrated in mostly non-overlapping regions of the state space. We consider them consistent when their high-probability regions largely overlap, even if their central tendencies differ. We also regard the distributions as consistent when a broad high-probability region of one distribution contains a substantially narrower region of the other. This situation may arise frequently because the predictive distribution contains information accumulated from many previous spikes.

In the analyses here, "local" means that each diagnostic is evaluated once per spike. The filter updates with the likelihood of everything observed in a time bin, including neurons that remained silent, whereas each diagnostic compares the predictive distribution with the normalized likelihood of a single spike; bins without spikes therefore produce no diagnostic value. "Local" does not mean that the predictive distribution is restricted to recent data: it summarizes all observations through the preceding time bin via the recursive state-space filter. More generally, the same comparisons could use a filtering or smoothing distribution defined over another chosen window, including a window containing both past and future observations.

Local goodness-of-fit measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model. All three are implemented in the open-source Python package `statespacecheck`, which can be applied to the output of other state-space decoders (see Discussion and Code availability).

All three metrics are evaluated at the level of individual spike events. Let C be the number of recorded neurons, let $n_{k,c}$ be the spike count of neuron c in time bin k, with $\mathbf{n}_k = (n_{k,1}, \ldots, n_{k,C})$, let $N_k = \sum_c n_{k,c}$ be the total population count, and let $y_{k,j}$ denote the mark of the jth event: the neuron identity for a spike-sorted model or the waveform features for a clusterless model. For a spike-sorted model, the observation y_k is the count vector $\mathbf{n}_k$, whose events carry the marks $y_{k,1}, \ldots, y_{k,N_k}$, so the observation likelihood $p(y_k \mid x_k)$ is $p(\mathbf{n}_k \mid x_k)$. For conditionally independent Poisson counts, it can be factored as

$$p(\mathbf{n}_k \mid x_k) = \prod_{c=1}^{C} \text{Poisson}\big(n_{k,c}; \lambda_c(x_k)\Delta t\big) = \frac{\exp[-\Lambda(x_k)\Delta t]}{\prod_{c=1}^{C} n_{k,c}!} \prod_{j=1}^{N_k} \left[\lambda_{y_{k,j}}(x_k)\Delta t\right], \tag{1}$$

where Δt is the bin width and $\lambda_c(x_k)$ is the conditional firing rate of neuron c in the bin. Its expected count is $m_c(x_k) = \lambda_c(x_k)\Delta t$. The total population rate is $\Lambda(x_k) = \sum_c \lambda_c(x_k)$, giving a total expected count of $\Lambda(x_k)\Delta t$. Equation (1) separates the exposure factor $\exp[-\Lambda(x_k)\Delta t]$, which is shared by the whole bin, from one event factor $\lambda_{y_{k,j}}(x_k)\Delta t$ for each observed event.

For the spike-sorted models analyzed here, HPD overlap and KL divergence compare the one-step predictive density

$$P_k(x_k) = p(x_k \mid y_{1:k-1})$$

with the normalized single-event likelihood obtained from that event's factor in Eq. (1):

$$Q_{k,j}(x_k) = \frac{\lambda_{y_{k,j}}(x_k)}{\int \lambda_{y_{k,j}}(u)\, du}. \tag{2}$$

The common factor Δt cancels in this normalization. $Q_{k,j}$ contains only the likelihood factor

attributable to the selected event. It deliberately excludes the bin-level exposure term, silent neurons, and total spike count, which cannot be assigned to one particular event. Those aspects of fit require a complementary count- or time-rescaling diagnostic. If several spikes occur in one bin, each event is evaluated separately against the same one-step predictive density P_k. For a clusterless model, the analogous $Q_{k,j}$ is obtained by normalizing the marked event-intensity density over the state space.

Overlap between HPD regions

One way to assess the consistency between the predictive distribution and the observation likelihood is to examine the overlap between their highest probability-density (HPD) regions. We refer to the resulting measure as HPD overlap. A substantial intersection indicates that the distributions assign high density to some of the same state values. Minimal or no overlap indicates disagreement under this criterion. This metric provides an interpretable geometric summary of local consistency between the distributions.

We define the overlap between the HPD regions of the predictive distribution and the normalized single-event likelihood as

$$\mathrm{HPDOverlap}(P_k, Q_{k,j}) = \frac{\left|\mathrm{HPD}_{P_k} \cap \mathrm{HPD}_{Q_{k,j}}\right|}{\min\left\{\left|\mathrm{HPD}_{P_k}\right|, \left|\mathrm{HPD}_{Q_{k,j}}\right|\right\}}, \tag{3}$$

where HPD_{P_k} and $\mathrm{HPD}_{Q_{k,j}}$ are the highest probability-density regions of the predictive density and the normalized single-event likelihood, respectively, and $|\cdot|$ denotes the volume of the region. The numerator is the size of the overlap between these regions, and the denominator is the size of the smaller of the two HPD regions. This ratio is the Szymkiewicz–Simpson overlap coefficient applied to the two HPD regions. Unlike the Jaccard index, which divides by the size of the union and therefore penalizes any difference in region size, the overlap coefficient equals 1 whenever one region is nested inside the other, which is exactly the nesting-tolerant behavior our consistency criterion requires. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap and 1 indicating that one region is entirely contained within the other. For

example, if the likelihood region is wider and therefore less informative than the predictive region but completely contains it, then the HPD overlap will be 1, suggesting that the two distributions are consistent under this criterion. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting substantial differences between the ranges of state values inferred from the two distributions. Throughout, we compute 95% HPD regions.

Fig. 2a illustrates the computation of the HPD overlap. We compute high-density regions for the predictive distribution (top) and the normalized single-event likelihood (middle), then divide the volume of their intersection by the volume of the smaller region (bottom).

Predictive checks

The HPD overlap compares two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. This approach has been used before in other contexts [12–14] and has the advantage that it allows for the computation of p-values for individual observations, which are easy to interpret [15, 16].

Predictive checks for spiking data can be challenging to implement because, within sufficiently short intervals, a spike is typically less probable than no spike. One possible approach is to consider spike patterns over longer fixed intervals in which multiple spikes are expected. Here, we take a simpler approach of focusing only on spike times and measuring the probability that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

Using the event notation introduced above, the relevant observation distribution is the distribution of the spike's mark for an event drawn from the model-implied marked point process. Because the diagnostic is evaluated once for each spike event, its predictive reference distribution corresponds to selecting an event rather than selecting a time bin that contains one or more events. Under the one-step predictive distribution P_k, the expected number of events in the bin associated with a state

region dx_k is $\Lambda(x_k)\Delta t\, P_k(x_k)\, dx_k$. The latent state associated with a randomly selected predicted event therefore has the event-weighted distribution

$$P_k^{\text{event}}(x_k) = \frac{\Lambda(x_k)P_k(x_k)}{\int \Lambda(u)P_k(u)\, du}. \tag{4}$$

At states where $\Lambda(x_k) > 0$, the probability that the event came from neuron c is

$$\pi_c(x_k) = \frac{\lambda_c(x_k)}{\Lambda(x_k)}. \tag{5}$$

States where $\Lambda(x_k) = 0$ receive zero mass under P_k^{event}. Combining the event-weighted state distribution with the conditional cell-identity distribution gives the predictive probability of cell identity:

$$f_{\text{pred},k}(c) = \int \pi_c(x_k) P_k^{\text{event}}(x_k)\, dx_k = \frac{\int \lambda_c(x_k)P_k(x_k)\, dx_k}{\sum_{c'=1}^{C} \int \lambda_{c'}(x_k)P_k(x_k)\, dx_k}. \tag{6}$$

For a clusterless model, let $\lambda(y, x_k)$ be the event rate per unit mark volume at waveform mark y, so that $\Lambda(x_k) = \int \lambda(w, x_k)\, dw$ is the total event rate and $\Lambda(x_k)\Delta t$ is the total expected event count in the bin. The corresponding predictive mark density is

$$f_{\text{pred},k}(y) = \frac{\int \lambda(y, x_k)P_k(x_k)\, dx_k}{\int \Lambda(x_k)P_k(x_k)\, dx_k}. \tag{7}$$

Equation (6) is the discrete spike-sorted form of Eq. (7). Let $\widetilde{y}_{k,j}$ denote a replicated mark distributed according to $f_{\text{pred},k}$. We compute the probability that the observed spike is at least as likely as this replicate:

$$p_{k,j} = \Pr_{\widetilde{y}_{k,j} \sim f_{\text{pred},k}} \left(\log f_{\text{pred},k}(y_{k,j}) \geq \log f_{\text{pred},k}(\widetilde{y}_{k,j}) \right). \tag{8}$$

For the spike-sorted models considered here, $\widetilde{y}_{k,j}$ takes one of a finite set of values—the identities of the recorded neurons. We compute Eq. (8) exactly as

$$p_{k,j} = \sum_{\substack{c \in \{1,\ldots,C\}: \\ f_{\text{pred},k}(c) \leq f_{\text{pred},k}(y_{k,j})}} f_{\text{pred},k}(c), \tag{9}$$

which sums the predictive probabilities of all cell identities at least as unlikely as the observed identity. The same probability could instead be approximated by Monte Carlo for a continuous or otherwise intractable mark space: draw x_k from the event-weighted distribution P_k^{event}, draw a replicated mark conditional on that state and an event, and compute the fraction whose predictive density is less than or equal to that of the observed mark. Equivalently, one may simulate from the full state-dependent point process and retain the marks of generated events. Because $p_{k,j}$ ranks the observed spike against this predictive distribution of possible spikes, we refer to it as the *rank-based predictive p-value*.

Fig. 2b illustrates the sampling interpretation of a predictive p-value. Values of the state are sampled from the event-weighted predictive distribution (top), each of which is used to sample a mark that could accompany the event (middle). The predictive density of the observed mark is then ranked against the densities of the replicated marks (bottom). In the schematic, the total expected event count is constant across states, so the event-weighted state distribution equals the one-step predictive distribution. Although this diagram shows the Monte Carlo construction, the spike-sorted results in this paper use the equivalent exact finite sum. High p-values suggest that the observed spike is consistent with the predictive distribution of spikes, and a p-value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

While HPD overlap explicitly assesses our definition of consistency in the state space, the predictive check probes a related notion in the observation space through the marginalization in Eq. (7). When the observation model is sufficiently informative about the state, we expect large p-values when the predictive distribution and likelihood match, but also when either distribution is narrower than the other, as long as they point to overlapping regions of the state space. This correspondence is not guaranteed: marginalization can map distinct state distributions to similar mark distributions, so a high predictive p-value does not by itself establish that the state distributions overlap. In the common case that the predictive distribution of the state is narrow relative to the likelihood, the predictive distribution of observations will be concentrated on the spikes that occur

at the confidently predicted location, and the observed spike will have appreciable likelihood at that location. The observed spike is among those that the model anticipates and should therefore lead to a high p-value. In the less common case that the likelihood is narrow relative to the predictive distribution of the state, the predictive distribution of the observations will also be broad and will assign appreciable probability to a wide range of spikes. The observed spike is likely to fall within this anticipated range, so the p-value should also be high. A low p-value tends to arise when the observed spike is specifically unlikely relative to what the model anticipates, which corresponds to the likelihood concentrating in a region the accumulated past deems improbable.

KL divergence

Finally, one of the best-known measures of discrepancy between probability distributions is the Kullback–Leibler (KL) divergence [17], defined in this case as the expected excess logarithmic loss incurred when the normalized single-event likelihood is used to approximate the predictive distribution, with the expectation taken with respect to the predictive distribution:

$$D_{\mathrm{KL}}(P_k \,\|\, Q_{k,j}) = \int P_k(x_k) \log\!\left(\frac{P_k(x_k)}{Q_{k,j}(x_k)}\right) dx_k, \tag{10}$$

where P_k and $Q_{k,j}$ are the distributions defined above. This definition matches the per-spike likelihood used for the HPD overlap and averaged within each time bin in the likelihood rows of Figs. 3 and 4.

Fig. 2c illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized single-event likelihood as functions of x_k. The middle panel shows their pointwise log-density ratio, $\log\big(P_k(x_k)/Q_{k,j}(x_k)\big)$. The bottom panel shows this ratio weighted by $P_k(x_k)$, which, when integrated, gives the value of the KL divergence.

The KL divergence is the most familiar way to compare two distributions, and we include it as a natural point of reference. It is an asymmetric measure, meaning that, in general, $D_{\mathrm{KL}}(P \,\|\, Q) \neq D_{\mathrm{KL}}(Q \,\|\, P)$. We could also compute the KL divergence from the likelihood to the predictive

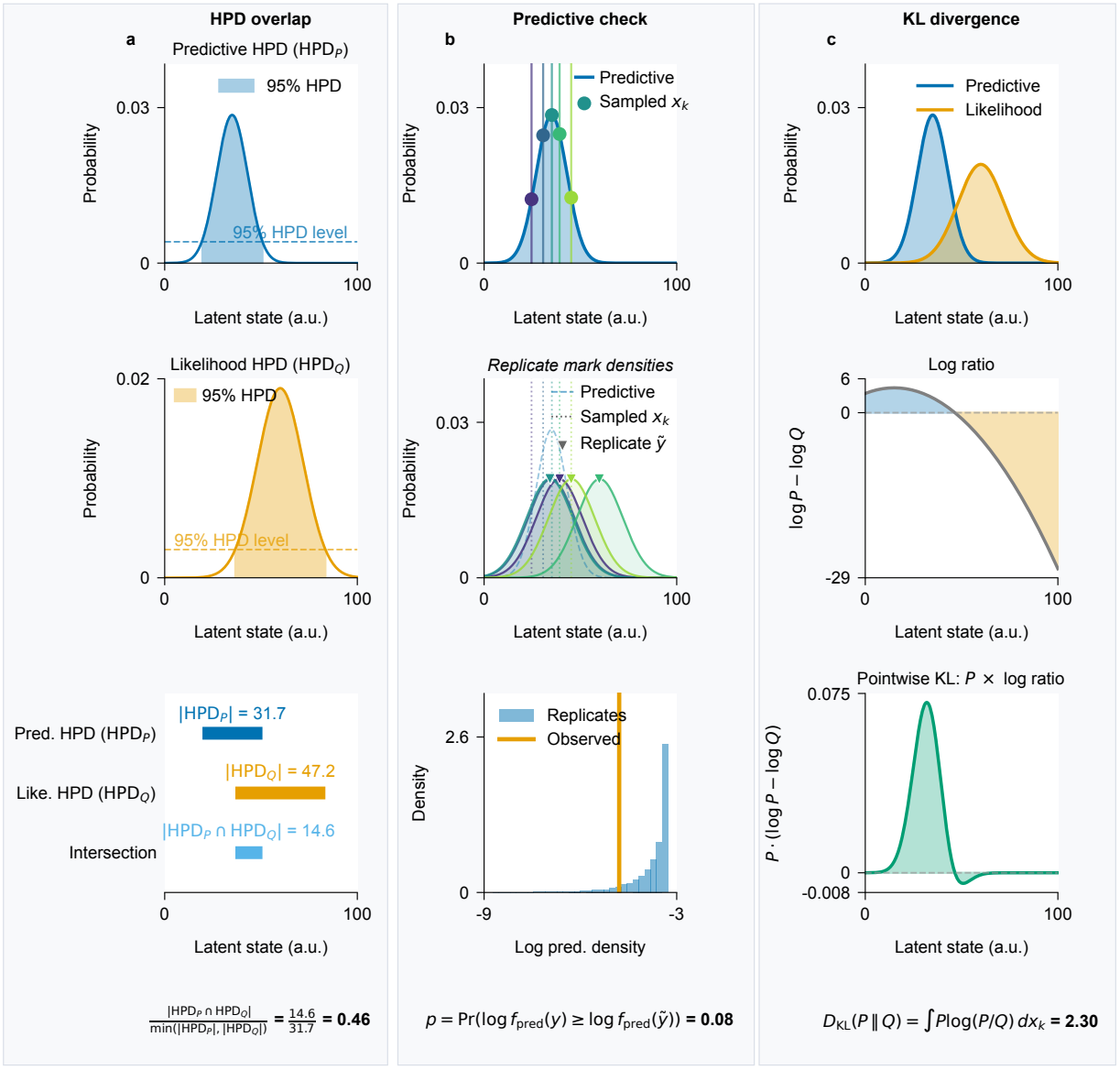


Fig 2. Computation of the three local goodness-of-fit metrics. **(a)** For HPD overlap, high-density regions are computed for the predictive distribution (top) and the normalized single-event likelihood (middle). The volume of their intersection is then divided by the volume of the smaller region (bottom). **(b)** The conceptual sampling construction for the predictive check: state values are sampled from the event-weighted predictive distribution (top), and each sampled state is used to generate a mark that could accompany the event (middle). The total expected event count is constant across states in this schematic, so the event-weighted and ordinary one-step predictive distributions coincide. The metric is the probability that the log predictive density of the observed mark is greater than or equal to that of a replicated mark (bottom); for the finite spike-sorted observation spaces analyzed here, this probability is evaluated exactly rather than by sampling. **(c)** For the KL divergence, the predictive distribution (blue) and the normalized single-event likelihood (orange) are compared (top) by computing the difference in their log densities (middle) and taking its expectation with respect to the predictive distribution (bottom).

distribution, $D_{\mathrm{KL}}(Q \parallel P)$, or the average of these two divergences, $\frac{1}{2}(D_{\mathrm{KL}}(P \parallel Q) + D_{\mathrm{KL}}(Q \parallel P))$, but both alternatives lead to a common issue, namely that the KL divergence can become arbitrarily large when its first argument places appreciable probability where its second argument assigns very little. Thus, $D_{\mathrm{KL}}(Q \parallel P)$ can become large when an uninformative spike produces a broad likelihood. Choosing $D_{\mathrm{KL}}(P \parallel Q)$ avoids that particular behavior but raises a complementary one: the divergence will grow large whenever the predictive distribution is broad relative to the likelihood, which can occur, for example, when informative spikes are sparse. We will see in the simulation and data analysis below that this behavior flags many spikes, even when their information is consistent with the accumulated information from past spiking.

Simulation study

Validating local goodness-of-fit metrics on real neural data is fundamentally limited: when the truth is unobserved, a metric that flags a moment of model failure cannot be distinguished from one that flags a moment of unusual neural activity. We therefore first evaluate the three diagnostics—HPD overlap, the rank-based predictive p-value, and KL divergence—in a controlled benchmark in which each perturbation is known by construction. The benchmark simulates a hippocampal place-cell population on a one-dimensional track and runs a Bayesian state-space decoder over the full session. Three windows deliberately perturb either the generative process or the decoder's assumptions; two additional controls present unusual neural activity that the decoder's observation model describes correctly. Together, these conditions distinguish model misfit that leads to decoding errors from differences in distribution shape that do not imply conflicting state information. The benchmark thus tests whether the diagnostics identify periods when current spiking conflicts with the model's prediction based on past spiking.

Generative model

We apply a Bayesian filter to a simulated 32-second session containing three misfit windows, three clean-recovery windows, and two controls whose spikes the decoder's observation model describes correctly.

Track and observation model. The state space is a one-dimensional linear track of length $L = 100$ arbitrary units (a.u.) discretized at unit spacing, $x \in \{0, 1, \ldots, 100\}$. The simulated population consists of eleven place cells with Gaussian place fields centered at $\mu_c \in \{0, 10, \ldots, 100\}$ and shared standard deviation $\sigma_{\text{pf}} = 10$ a.u. The expected spike count of cell c per 1 ms time step at position x is

$$m_c(x) = \lambda_c(x)\Delta t = \alpha\, \phi\Big(x \mid \mu_c, \sigma^2_{\text{pf}}\Big), \tag{11}$$

where $\Delta t = 1$ ms, $\phi(\cdot \mid \mu, \sigma^2)$ is the Gaussian density, and $\alpha = 5$ is a scale parameter for the expected count. The peak expected count is $\alpha/(\sigma_{\text{pf}}\sqrt{2\pi}) \approx 0.20$ spikes per step, corresponding to a peak rate of approximately 200 Hz. Spike counts are drawn independently across cells at each step, $n_{k,c} \sim \text{Poisson}(m_c(x_k))$. Except during the replay control described below, x_k is both the animal's physical position and the position represented by the simulated ensemble.

Five additional cells form a sparse population of narrow fields clustered near $x = 30$ a.u. (centers spread over 28.5–31.5 a.u., each with standard deviation 2 a.u.). Each fires as an independent Poisson process. Its per-cell peak rate is 0.01 Hz before the final sparse-population control and 1 Hz during it; with five cells the aggregate rate stays low (a few spikes per second), so the spikes remain temporally isolated. The decoder uses these same phase-specific rates, so recruiting these cells does not itself create an observation-model mismatch.

Position dynamics. Except where noted below, the animal's position evolves as a Gaussian random walk with step standard deviation $\sigma_{\text{pred}} = 0.5$ a.u. The trajectory is reflected at 0 and 100 a.u. throughout the session, starts at $x_0 = 0$, and is continuous across phases: the final position of each phase becomes the initial position of the next.

Phase schedule. The 32-second simulation (32,000 steps) is partitioned into eight contiguous phases: an opening baseline, three misfit windows, three clean-recovery windows, and a final sparse-population control. Each misfit window perturbs the observation model $p(y_k \mid x_k)$ or the state-transition model $p(x_k \mid x_{k-1})$ by altering either the generative process or the decoder's assumptions:

Remap: During steps 6,000–10,000, the decoder-side observation model is perturbed. The decoder evaluates per-spike likelihoods against a globally remapped set of place fields: the eleven place-field centers are scrambled by a fixed random permutation (a derangement in which every field is displaced by at least three place-field spacings), modeling global remapping to statistically independent locations. Both the decoder's posterior update and its per-spike diagnostics use this remapped likelihood—the diagnostics are evaluated against the model the decoder actually uses, not the true generative fields—so the misfit surfaces from the scramble's spatial incoherence rather than from any oracle knowledge of the correct fields. Spike generation is unchanged.

History-dependent firing: During steps 14,000–18,000, the generative-side observation model is perturbed. Spike counts are generated with each cell's Poisson mean set to zero for 1 ms after a step in which it spiked, followed by a 2 ms–10 ms burst window with a threefold rate increase, matching the rough phenomenology of CA1 pyramidal-cell refractory and burst windows; the decoder still assumes independent Poisson firing. Because the burst window outlasts the suppressed step, this modulation also raises each cell's mean firing rate above the Poisson rate the decoder assumes. The per-spike spatial likelihood is unchanged, so the mismatch lies in the temporal joint distribution and overall rate of the spike train, not in the spatial information each spike carries.

Drift: During steps 22,000–26,000, the generative-side state-transition model is perturbed. The trajectory is generated with persistent velocity, $v_k = 0.88\, v_{k-1} + \eta_k$ and $x_k = x_{k-1} + v_k$ with $\eta_k \sim \mathcal{N}(0, \sigma^2_{\text{pred}})$; the decoder continues to assume a memoryless random walk with step

standard deviation σ_{pred}.

The three intermediate clean-recovery windows (steps 10,000–14,000; 18,000–22,000; 26,000–30,000) restore both the generative process and the decoder to the baseline configuration, apart from the replay event embedded in the second (described below) and the approach at the end of the third, providing out-of-sample well-specified comparison windows. They are summarized in Fig. 3b but are not used to set the diagnostic thresholds, which are derived from the opening baseline. During the last second of the final clean-recovery window, the animal moves at constant speed to $x = 30$ a.u. so that the sparse-population control begins without an abrupt position jump; these steps remain in the pooled well-specified column.

Replay control. Embedded in the second clean-recovery window (steps 19,000–21,000) is a replay event. For this control, we distinguish the animal's physical position, z_k, from the internally represented position, x_k, that drives the place-cell spiking. The animal is immobile, so z_k remains fixed, while x_k moves at constant speed from its starting point to the farther end of the track and back. During the sweep the place cells fire at an elevated rate, with $\alpha = 20$ in Eq. (11) (a peak rate of approximately 800 Hz). The decoder uses these elevated rates, so its observation model matches the replayed spikes. Its random-walk transition does not describe the deterministic sweep, but each step of the sweep is small relative to σ_{pred}, so the decoder tracks x_k—its decoded position departs markedly from the animal's fixed physical position z_k while the predictive distribution and each spike's likelihood remain in agreement throughout the sweep. Because the diagnostics assess the internal consistency of the decoded state rather than its correspondence to physical position, this large decoded-to-physical separation is correctly left unflagged: it reflects a change in what the ensemble represents, not a failure of the model. The replay steps are excluded from the pooled well-specified comparison column and are scored as their own column in Fig. 3b.

Sparse-population control. During steps 30,000–32,000, the animal is immobile at $x = 30$ a.u. and the ordinary place-cell ensemble falls silent (its expected count is set to zero), as can

occur during immobility. The clustered sparse-population cells increase to their 1 Hz peak rate and fire intermittently. The decoder's observation model is correctly specified: it knows that the ordinary ensemble is silent and knows the active sparse-population rates. Its random-walk transition, by contrast, does not describe the immobile animal. Because that prior expects the animal to move, the predictive distribution gradually spreads in the intervals between isolated spikes; each sparse-population spike then supplies a narrow likelihood centered inside that prediction. The two distributions therefore remain consistent by the HPD-overlap and predictive-check criteria, while their large difference in concentration can elevate KL divergence.

With this schedule fixed, each diagnostic can be evaluated against the known model faults and the two controls.

Simulation results

Fig. 3a shows a realization of the simulation, the decoding results, and the three local goodness-of-fit metrics. The top panel shows the physical position of the simulated rat in magenta and the predictive distribution in black. The regions with white backgrounds are the opening baseline and the clean-recovery windows; colored backgrounds indicate the three misfit windows and two controls. The second panel shows the mean normalized per-spike likelihood as a function of position in each spike-containing time bin; when multiple spikes share a bin, their normalized likelihoods are averaged for display only. The diagnostics themselves are computed for each spike. The third panel shows the spikes from each of the sixteen simulated neurons. Since these are sorted by their place-field centers, the pattern of spiking aligns with the likelihood as a function of position. Note that the remap window is the one in which the predictive distribution and the spike likelihoods most clearly disagree about the animal's location. During the replay control, by contrast, the predictive distribution departs from the physical position yet continues to agree with the spike likelihoods. The sparse-population control is visibly different: after the ordinary ensemble falls silent, isolated spikes appear only in the clustered sparse-population cells.

The lower three panels of Fig. 3a show, from top to bottom, the values of the HPD overlap,

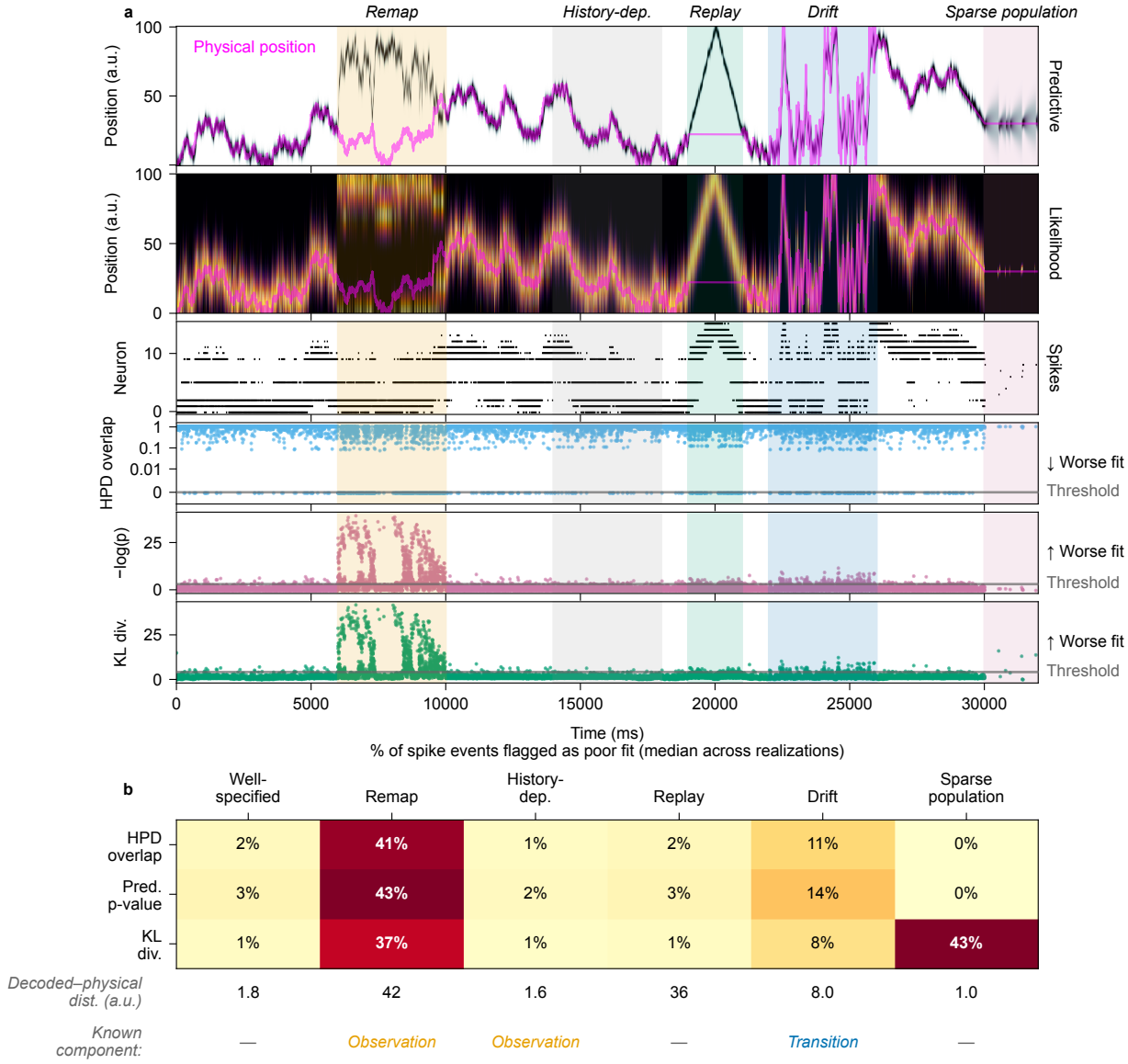


Fig 3. Per-spike goodness-of-fit diagnostics for the simulation study. **(a)** Top row: the predictive distribution over time (1 ms per step) and position, with the animal's physical position in magenta; shading marks the three misfit windows and two controls. Second row: the mean normalized per-spike likelihood in each spike-containing bin (averaged for display only). Third row: the spike raster, with neurons sorted by place-field peak position. Bottom rows: HPD overlap (symmetric-log axis), $-\log(p)$ of the rank-based predictive p-value, and KL divergence for each spike event, with flag thresholds. **(b)** Median percentage of spike events flagged in each phase across 100 realizations, with the clean-recovery windows (excluding the replay event) pooled into the well-specified column. HPD and KL thresholds come from the pooled opening baseline (steps 0–6,000); the predictive p-value uses a fixed threshold of 0.05. The "Decoded–physical dist." row gives the median absolute distance (a.u.) between the decoded position (filtered posterior mean) and the animal's physical position. The bottom row names the perturbed component (*Observation* or *Transition*); a dash marks well-specified and control phases.

the rank-based predictive p-value (as $-\log(p)$), and KL divergence for each spike. In each panel, a horizontal line marks the threshold used to flag poor fit. Because low HPD overlap indicates disagreement, a spike is flagged when its value is at or below the 1st percentile of values from the opening baseline (steps 0–6,000) pooled across 100 independent simulation realizations. For the rank-based predictive p-value, a spike is flagged when $p \leq 0.05$ (equivalently, $-\log(p) \gtrsim 3$), a fixed threshold rather than a baseline-derived one. Because high KL divergence indicates disagreement, a spike is flagged when its value is at or above the 99th percentile of values from the same pooled opening-baseline distribution. Pooling across realizations stabilizes the baseline-derived HPD and KL thresholds, which would otherwise be noisy when estimated from a single run. Because many spikes have HPD-overlap values tied at the threshold, the HPD rule flags 1.4% of opening-baseline spikes, more than the nominal 1st-percentile fraction.

We see that all three metrics agree that there is a lack of consistency during the remap window. There, the decoder evaluates each spike against the scrambled place fields, so successive spikes, which the true fields generate from neighboring positions, have likelihoods centered at unrelated locations. Each new spike's likelihood therefore tends to fall outside the region supported by the predictive distribution built from the preceding spikes. During the history-dependent-firing window, the estimated decoding distribution becomes narrower, because the bursting cells fire more often, but remains consistent with the spike likelihoods, and all three metrics flag no more spikes than in the well-specified windows. The replay control also stays near the baseline flag rate, though for a different reason: although the decoded position sweeps far from the animal's physical location, the predictive distribution tracks the replayed trajectory and remains consistent with each spike's likelihood. During the drift window, when the decoder omits the persistent velocity, the actual movement trajectory has notably increased speed. The decoder only partially keeps up, so its predictive estimates lag behind the physical location; this lag biases the predictive spiking distribution toward neurons consistent with the lagged position. As a result, the diagnostics flag a modest percentage of spikes during the drift window—most clearly the rank-based predictive p-value—though far less than during the remap window. During the sparse-population control, the

ordinary ensemble is quiet and the prediction broadens naturally between intermittent spikes from the clustered cells. Each such spike concentrates inside the range of positions already supported by the prediction. HPD overlap and the rank-based predictive p-value therefore classify the two distributions as consistent, whereas KL divergence responds to their large difference in concentration. This familiar sparse-firing regime illustrates why distributional difference need not imply conflicting state information.

Fig. 3b shows the percentage of spikes flagged by each metric in each phase, summarized by the median across 100 independent realizations of the simulation. The median percentage flagged during the remap window is substantial for all three metrics (37–43%), demonstrating that all three methods can identify inconsistent spiking information leading to incorrect state estimates; the percentage nonetheless varies from realization to realization because it depends on how much of the scrambled map the simulated trajectory visits. All three metrics flag only 1–2% of spikes during the history-dependent-firing window. The replay control is flagged at a low rate—a median of 1–3% across the three metrics, comparable to the well-specified windows—confirming that a large decoded-to-physical separation is not, by itself, registered as model misspecification. The drift window is flagged more often (8–14%), most notably by the rank-based predictive p-value, but still far less than during the remap window. In the sparse-population control, HPD overlap and the rank-based predictive p-value flag no spikes in the median realization, whereas KL divergence flags a median of 43%. This is consistent with the realization shown in Fig. 3a, where HPD overlap and the rank-based predictive p-value registered the spikes and the prediction as consistent, while KL divergence responded to the difference in concentration between the predictive distribution and the likelihood, even though the two carry consistent information about the state process.

The "Decoded–physical dist." row of Fig. 3b compares these flag percentages with the median absolute distance between the decoded position (filtered posterior mean) and the animal's physical position. Outside replay, this distance is a decoding error. In the well-specified windows, the median absolute error of the filtered posterior mean is 1.8 a.u., less than one-fifth of a place-field width. Remapping raises it to 42 a.u., matching the high flag percentages. History-dependent firing

leaves it essentially unchanged (1.6 a.u.), consistent with its low flag percentages: this misfit alters the temporal structure and rate of the spike train without displacing the spatial information each spike carries. The drift window, by contrast, exposes a limitation of event-level diagnostics. The median error rises to 8.0 a.u., more than four times the well-specified value, because the filtered posterior lags the fast-moving position, yet only 8–14% of spikes are flagged. Because each spike is compared with a predictive distribution that the preceding spikes have already pulled toward the lagged estimate, a persistent lag produces a modest per-spike disagreement rather than a large one. During the replay control, the decoded-to-physical distance is large by construction (36 a.u.) because the decoder is tracking the represented trajectory, so this separation reflects the replayed trajectory rather than a decoding error. The sparse-population control is decoded accurately (1.0 a.u.) even though KL divergence flags 43% of its spikes, confirming that those flags do not correspond to decoding errors.

Real hippocampal data analysis

Data collection and behavioral task

We applied the local goodness-of-fit metrics to units recorded from the rat hippocampus during one recording epoch, in which the animal performed a spatial bandit task on a maze with three radially arranged Y-shaped foraging patches extending from a central point. A schematic of the maze is shown in the right inset in Fig. 4. The details of this task and the data collection methods are described by Comrie et al. [18]. Briefly, the rat was trained to run between six reward ports located in three Y-shaped "patches". Each port was assigned a reward probability of 0.2, 0.5, or 0.8, such that one patch was best on average. Reward probabilities changed periodically after a preset time or trial limit. Recordings were made with 22 tetrodes advanced to CA1; two additional tetrodes, one in each hemisphere, served as local references in the corpus callosum.

For the present analysis, we sorted the continuous 30 kHz neural signal into putative single units with MountainSort4 [19]. In the MountainSort4 pipeline, the signal was band-pass filtered

and whitened during preprocessing; negative-going events were then detected on each tetrode at a threshold of 3 in whitened units, with a minimum inter-event interval of 10 samples (≈0.33 ms). Units were retained from automated curation only, without manual cluster curation. Because curation was automated, some units may contain multiunit activity or represent incompletely merged clusters; this does not affect our purpose here, which is to demonstrate the diagnostics rather than to characterize individual neurons.

For decoding, we linearized the animal's position by projecting it to the nearest point of a skeletonized track, discretized the resulting linear position into spatial bins of approximately 2 cm, and binned spikes at 2 ms resolution. We fit both decoders with `non_local_detector` version `0.6.10.dev214+g956fdccaf`. They used the same sorted-spikes kernel-density observation model, with positional bandwidth `position_std`= $\sqrt{12.5} \approx 3.5$ cm, and initialized position uniformly over the track-interior bins. The Continuous model assigned probability one to its sole dynamics mode and used a zero-mean Gaussian random-walk position transition with `movement_var`= 6.0 cm^2. The Continuous–Fragmented model initialized the Continuous and Fragmented modes with probabilities $(0.5, 0.5)$ and used the mode-transition matrix

$$\begin{pmatrix} 0.98 & 0.02 \\ 0.02 & 0.98 \end{pmatrix},$$

with rows and columns ordered as (Continuous, Fragmented). Its Continuous-to-Continuous position transition used the same random walk as the Continuous model; the other three mode transitions used a uniform distribution over track-interior position bins.

The Continuous–Fragmented decoder represented the joint state of discrete dynamics mode and position, with the same position-dependent observation model in both modes. For all three diagnostics, we marginalized its one-step predictive distribution over dynamics mode and compared the resulting distribution over position with one copy of the shared observation likelihood. The diagnostics for both decoders were therefore defined on the same position grid. The rank-based predictive p-value used the same fixed threshold as in the simulation, flagging $p \leq 0.05$. For HPD

overlap, the recording has no designated clean baseline period from which to derive a percentile threshold, so we flagged values at or below a fixed threshold of 0.05.

Data analysis results

Fig. 4 compares decoding of these hippocampal spike data under two state-space models. The first assumes that the latent state moves continuously through space, as expected if it represents only the animal's physical location. During immobility, however, hippocampal representations can jump to distant locations and reactivate in sequences resembling past movement trajectories [20]. We therefore expect the Continuous model to exhibit local disagreement with the spiking data when the neural representation jumps. We compare this model with a Continuous–Fragmented model that permits discrete jumps in the hippocampal representation. We previously used a related model structure to capture replay and other nonlocal hippocampal representations [2].

Fig. 4a shows a brief segment decoded with the Continuous model while the animal is immobile. The rows follow the same order as in Fig. 3a. During this candidate replay event, the decoded representation transitions rapidly among maze locations while the animal remains at rest, as indicated by the magenta position trace. The Continuous model cannot track these rapid transitions, and its predictive distribution lags behind the information provided by the likelihoods of the observed spikes. The HPD overlap and rank-based predictive p-value diagnostics flag many spikes, and several spikes also have elevated KL divergence; KL divergence values are not dichotomized in this analysis.

The contrast between the Continuous and Continuous–Fragmented models is shown in panels a and b of Fig. 4. The likelihood of each spike is identical under the two models (second row), but the Continuous–Fragmented model decodes a rapidly moving state trajectory that the Continuous model cannot track. When the representation is assigned to the Fragmented mode, the predictive distribution becomes uniform or nearly uniform, allowing the likelihoods of subsequent spikes to produce a jump in the state estimate.

For HPD overlap (fourth row, blue), 38 of the 43 spikes in this window flagged under the Continuous model are no longer flagged under the Continuous–Fragmented model. Most spikes in

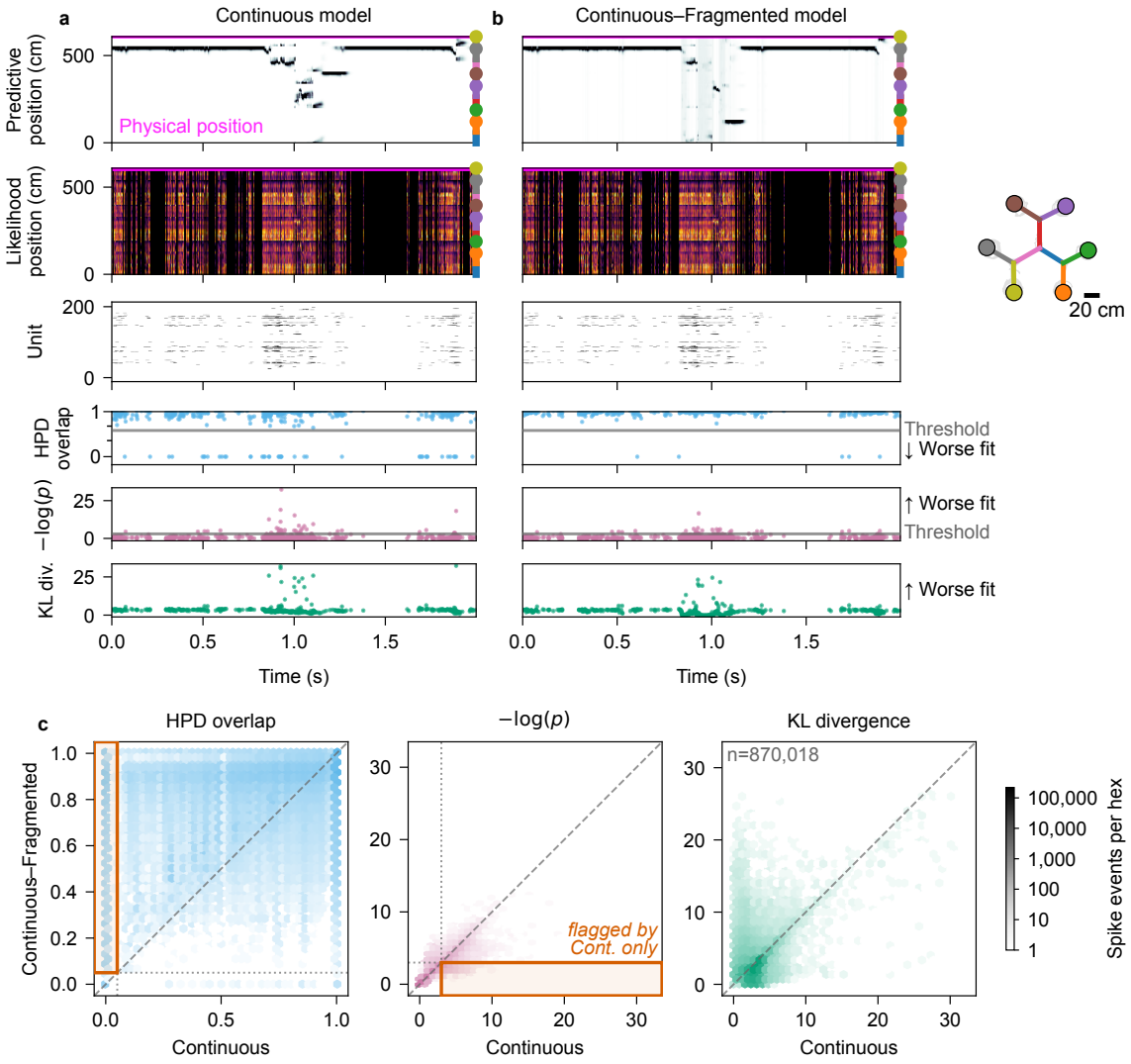


Fig 4. Per-spike goodness-of-fit diagnostics applied to a hippocampal recording during navigation on a multi-arm maze (203 sorted units, 182 of which fired in this epoch; data from Comrie et al. [18]), evaluated under a Continuous and a Continuous–Fragmented state-space model. The inset shows the two-dimensional maze with the animal's trajectory in gray; colored dots mark the six reward ports, and segment colors match the bars beside rows 1–2. **(a, b)** A window containing a candidate replay event, decoded with the Continuous (a) and Continuous–Fragmented (b) models; time is relative to the start of the window. Row 1: the one-step predictive distribution over linearized position, with the animal's physical position overlaid. Row 2: the mean normalized per-spike likelihood in each spike-containing bin (averaged for display only), identical under both models. Row 3: the spike raster, with units sorted by place-field peak position. Rows 4–6: HPD overlap (symmetric-log axis), $-\log(p)$, and KL divergence for each spike event. Horizontal lines mark the flag thresholds, 0.05 for HPD overlap and $-\log(0.05) \approx 3$ for the predictive p-value; KL divergence has none. **(c)** Per-spike diagnostic values across the recording epoch under the Continuous (horizontal) and Continuous–Fragmented (vertical) models; the dashed line is the identity. Dotted lines mark the flag thresholds; outlined boxes contain spikes flagged only under the Continuous model. Color gives the number of spike events on a shared logarithmic scale; n is the number of spike events in each panel.

the window (457 of 649) also have lower $-\log(p)$ values under the Continuous–Fragmented model than under the Continuous model (fifth row, pink). Under the Continuous model, the predictive distribution can concentrate at the wrong location, making the observed spike very unlikely. During replay events, the Continuous–Fragmented model instead produces a broad predictive distribution. Although the observed spike can remain unlikely under this broad distribution, it is less unexpected than under the concentrated Continuous-model prediction. For KL divergence (bottom row, green), many spikes retain high values under the Continuous–Fragmented model. This pattern is consistent with the simulation results: predictive distributions that are broad relative to spike likelihoods produce larger KL values even when our operational criterion classifies those prediction–likelihood pairs as consistent.

Fig. 4c shows hexagonally binned density plots comparing the per-spike values of each local goodness-of-fit metric, for all 870,018 spike events in the recording epoch, under the Continuous and Continuous–Fragmented models; color indicates the number of spike events in each hexagon. The left panel shows HPD overlap in blue. Hexagons above the 45° line correspond to spikes with greater HPD overlap under the Continuous–Fragmented model. The dense vertical band at a Continuous-model value of 0 and Continuous–Fragmented values above 0.05 (within the outlined box) represents spikes for which the Continuous model's predictive distribution has no overlap with the spike likelihood, whereas the Continuous–Fragmented model's predictive distribution has nonzero overlap, so adding the Fragmented mode alleviates this disagreement for many spikes. There is comparatively little density at a Continuous–Fragmented value of 0 and Continuous-model values above 0.05, indicating that it rarely creates new disagreement. Of the 18,790 spikes flagged under the Continuous model, 17,289 (92%) are no longer flagged under the Continuous–Fragmented model (*rescued*), whereas 176 spikes are flagged only under the Continuous–Fragmented model (*newly flagged*).

The center panel of Fig. 4c shows the negative logarithm of the rank-based predictive p-value, $-\log(p)$, in pink. Hexagons below the 45° line correspond to spikes that are less unexpected under the Continuous–Fragmented model than under the Continuous model. The density band below this

line extends beyond a Continuous-model value of 3 (within the outlined box), corresponding to $p \leq 0.05$ because $-\log(0.05) \approx 3$. The corresponding region above the line, where the Continuous–Fragmented value exceeds 3, is substantially less dense. Thus, the rank-based predictive p-value identifies many spikes for which local disagreement is reduced under the Continuous–Fragmented model. Of the 33,954 spikes flagged under the Continuous model, 9,373 (28%) are rescued by the Continuous–Fragmented model, whereas 1,706 spikes are newly flagged.

The right panel of Fig. 4c shows KL divergence values in green. Hexagons below the 45° line correspond to spikes with lower KL divergence under the Continuous–Fragmented model. Most spike events (94%) lie below the line, concentrated at Continuous-model values of 2.7–3.5 (interquartile range). More strikingly, the 6% of spike events above the line have a median Continuous-model value of 1.5, whereas their Continuous–Fragmented values extend to 30. As in the simulation, these values occur when the Continuous–Fragmented model produces a broad predictive distribution after estimating the Fragmented mode. KL divergence therefore aligns less closely with the nesting-tolerant consistency criterion used for the other metrics in this setting.

Discussion

State-space modeling has become a powerful tool for describing the latent dynamics of neural systems. Validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the information carried forward from all preceding spiking activity by the state-space model. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

Two of the three metrics we examined, HPD overlap and the rank-based predictive p-value, behaved as a consistency measure should: they flagged spikes that conflicted with the prediction, but

not spikes whose likelihood lay within a broad prediction. The HPD overlap provides a geometric measure of consistency that remains interpretable when one high-density region contains another. The rank-based predictive p-value provides a direct statistical assessment of each observation. For spike-sorted data on a discretized state space, the p-value is exact and inexpensive, a finite sum over the recorded neurons, whatever the state-transition model; for continuous marks, as in clusterless decoding, or for other observation models whose predictive mark distribution cannot be enumerated, it must instead be estimated by Monte Carlo sampling, which is considerably more costly. The KL divergence is computationally simple but is sensitive to distribution tails and lacks a clear upper bound, making it difficult to set a threshold. More broadly, KL divergence is only one member of the larger family of f-divergences, which also includes the total variation distance and the Pearson χ^2 divergence [21]. Substituting another member of this family would emphasize different features of the prediction–likelihood mismatch, but because these measures all target distributional difference rather than the consistency we seek, KL's tendency to flag spikes that remain consistent with the model, as occurs whenever the predictive distribution is broad, would likely carry over to the family as a whole. This behavior is a liability for consistency assessment, but it could be useful in cases where the goal is to flag time points when the predictive distribution becomes uncertain relative to the information coming from individual spikes. When the goal is consistency in estimates of the state process, we recommend HPD overlap and the rank-based predictive p-value as primary diagnostic measures.

We illustrated the application of these metrics to simulated and real spike-sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state-space models. In particular, clusterless decoding has been shown to give lower decoding error and better-calibrated uncertainty than decoding from spike-sorted units [5]. The metrics, as defined in the Methods section, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state-space model up to that point in time.

These metrics make it possible to identify specific periods when a state-space model is more or

less consistent with the spiking data, and to turn that diagnosis into a concrete modeling decision. Consider a researcher decoding the hippocampal recording with the Continuous model alone. During the candidate replay event, the local diagnostics reveal that many observed spikes conflict with a state representation constrained to move continuously through space. This pattern motivates the hypothesis that the continuous transition model is too restrictive. Consistent with that hypothesis, adding a Fragmented mode that permits discrete jumps eliminates the HPD-overlap flag for 92% of spikes flagged under the Continuous model across the recording epoch (Fig. 4c). A single score for the whole epoch could rank the two models, but it would not identify which individual observations drive the difference. Here, the local diagnostics, alternative-model comparison, and scientific context together support a specific refinement of the model. More broadly, this diagnose-then-refine cycle enables a feedback loop between model development and assessment that can improve the quality of state-space models.

To make these diagnostics straightforward to apply in other studies, we provide them as an open-source Python package, `statespacecheck` (https://github.com/edeno/statespacecheck). For a spike-sorted decoder, the package needs only quantities that a grid-based decoder already computes: the one-step predictive distribution in each time bin, each neuron's firing rate as a function of the state, and the time bin and neuron of each spike. From these it returns the HPD overlap, the rank-based predictive p-value (evaluated exactly over the recorded neurons), and the KL divergence for every spike, along with the building blocks (the normalized single-event likelihood and the predictive distribution over neurons) and a helper for setting thresholds from a baseline period. For clusterless decoders, the package computes the same three diagnostics from the joint mark intensity $\lambda(y, x_k)$, approximating the predictive p-value by Monte Carlo as in Fig. 2b. The package does not depend on how the decoder was fit, so it can be applied to an existing model without modification. Because the diagnostics yield one value per spike, it is often most informative to inspect them alongside the decoded state. A desktop viewer distributed with the analysis code displays a full decoded recording epoch with the per-spike diagnostics; the user can scrub through time, select an individual spike to compare its likelihood with the predictive distribution at that time, view the population likelihood

against the predictive, filtered, or smoothed distribution, and, for the hippocampal recording, switch between the Continuous and Continuous–Fragmented models. We also provide a companion website (https://edeno.github.io/statespacecheck-paper/) where readers can explore how the diagnostics are computed and inspect the simulated and hippocampal examples.

These methods could be extended in several ways. For simplicity, we focused on comparing instantaneous information in the likelihood of each spike with aggregated information from the past contained in the one-step predictive density. A natural and powerful extension of these methods is obtained by replacing the one-step predictive density with any posterior density derived from the state-space framework. In particular, replacing the one-step predictive density with a smoothing density (e.g., from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification. Because filtering and smoothing densities also condition on the spike being assessed, that spike's own contribution would first need to be removed, for example by leave-one-out smoothing, so that the spike is not compared against a distribution it helped form.

These approaches also present challenges. First, the computational cost of these metrics increases as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and predictive densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. We also evaluated the predictive checks exactly by summing over the finite set of recorded-neuron identities. For models with continuous or otherwise high-dimensional observation spaces, the predictive check could instead be approximated by sampling from the predictive state and observation distributions, avoiding enumeration of every observation outcome; whether this is more efficient will depend on the model and the required Monte Carlo precision. Sampling-based estimators could likewise improve the tractability of KL divergence and HPD overlap in moderate dimensions, although the number of samples required for accurate estimates will still grow with dimension. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space.

Identifying subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to choosing time scales for assessing lack of fit. These methods allow us to focus on time scales as short as individual spikes, and in our place-cell decoding examples, individual spikes are sufficiently informative that we are often able to detect local model misfit at this time scale. In other neural systems, local model errors may be easier to detect over longer time scales. In this paper, we note that these goodness-of-fit measures can be extended to longer time intervals, but determining which time scales are most appropriate may require relying on prior knowledge about the system, or exploring the values of these metrics across multiple time scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our models of neural dynamics and deepen our understanding of how the brain encodes and processes information.

Ethical statement

The recording analyzed here was collected for a previous study [18]. All animal procedures were approved by the Institutional Animal Care and Use Committee at the University of California, San Francisco (protocol AN174991).

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The authors declare that they have no competing interests.

Data availability statement

The hippocampal recording analyzed here was originally reported by Comrie et al. [18] and is available on the Distributed Archives for Neurophysiology Data Integration (DANDI) Archive as dandiset 001942 (`https://dandiarchive.org/dandiset/001942`). The decoder inputs for the recording epoch analyzed in Fig. 4—the animal's position, the spike times of the sorted hippocampal units, and the track geometry—are archived on Zenodo (doi:10.5281/zenodo.23020757).

Code availability

The local goodness-of-fit diagnostics are implemented in the `statespacecheck` Python package, available at `https://github.com/edeno/statespacecheck` and installable from the Python Package Index (`pip install statespacecheck`); the analyses in this paper used version `0.3.1`, archived on Zenodo (doi:10.5281/zenodo.23011969). All simulation, analysis, and figure-generation code, together with the interactive viewer and the source of the project website (`https://edeno.github.io/statespacecheck-paper/`), is publicly available at `https://github.com/edeno/statespacecheck-paper` and archived on Zenodo (doi:10.5281/zenodo.23019296).

Author contributions

Conceptualization: UTE, SZ.

Methodology: UTE, SZ.

Software: SZ, ELD.

Formal analysis: SZ, ELD.

Investigation: SZ, UTE, ELD.

Data curation: AEC.

Writing – original draft: SZ, UTE, ELD.

Writing – review & editing: SZ, AEC, LMF, UTE, ELD.

Visualization: SZ, ELD.

Supervision: UTE.

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